NITYA MUNGALPARA

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PROFESSIONAL SUMMARY

Final-year Computer Science (Professional) student at RMIT with minors in AI & Machine Learning and Cloud Computing, and an AWS Certified Cloud Practitioner. Delivered an ERPNext e-invoicing compliance system for an industry client in Italy (High Distinction), coordinated a six-person Spring Boot and React platform through CI/CD to a live client demo, and trained a CNN that identifies cancerous cells in unseen patients' tissue (High Distinction). Python tutor and peer mentor at RMIT.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, C#, Bash, Prolog

Backend and Web: Spring Boot, REST APIs, React, Node.js, Frappe / ERPNext, Javalin, Thymeleaf, HTML, CSS

Cloud (AWS): VPC, EC2, Auto Scaling, Application Load Balancer, Lambda, API Gateway, DynamoDB, S3, SQS, SNS, IAM, CloudWatch, CloudFormation, Amplify

DevOps and Tooling: Docker, GitHub Actions CI/CD, Git, Linux, Flyway database migrations, Jira

AI and Machine Learning: PyTorch, CNNs, transformers (RoBERTa), LoRA, scikit-learn, pandas, NumPy, RAG, BM25 and dense retrieval, LLM-as-judge evaluation

Databases: MySQL, SQLite, MongoDB, Neo4j, database design (ERDs)

Practices: Agile, unit testing, code review, requirements elicitation, technical documentation, AI-assisted development (GitHub Copilot, Claude Code)

EDUCATION

Bachelor of Computer Science (Professional), RMIT University, Melbourne 2024 - 2026

Minors: Artificial Intelligence & Machine Learning; Cloud Computing

Coursework completes November 2026; the final year of the Professional degree is a 12-month industry placement in 2027.

High Distinctions: Programming Project 1 (industry client); Algorithms and Analysis; Introduction to Cyber Security; Cloud Security; Cloud Developing; Software Engineering Fundamentals; Programming Studio 1. HD-graded assessments in Machine Learning (cancer cell classification) and Deep Learning (gesture recognition CNN, 27.7/30).

CERTIFICATIONS

- AWS Certified Cloud Practitioner, Amazon Web Services
- Quantum Computing School (RMIT), certified program; project on noise mitigation using Bayesian optimisation
- Working with Children Check (Victoria)

EXPERIENCE

Backend Developer and Documentation Lead, Industry Client Project SD Companies, Italy | Mar 2026 - Jun 2026

- Delivered a production ERPNext system for a real external client in a four-person team through RMIT (Programming Project 1), meeting the client regularly; graded High Distinction, with positive feedback from the client, mentor and course coordinator.
- Built a pre-submission validation layer checking outbound invoices against Italy's mandated FatturaPA standard and the rules enforced by the Sistema di Interscambio (SDI), rejecting non-compliant invoices with specific, actionable reasons identifying the offending record and value.
- Developed a customisable CRM-style sales pipeline on the Frappe framework with role-based access control and workflow automation.
- Elicited the client's taxation requirements and translated them into system behaviour; led the technical documentation and requirements artefacts the client and team worked from; tracked tasks and deadlines in Jira.

Software Team Member, RMIT Rover Team RMIT University | May 2026 - Present

- Embedded and control software for a student-built rover (Raspberry Pi, Pi Pico, Arduino, sensors, gimbal cameras) in a cross-disciplinary team using Git.

Python Programming Tutor RMIT University | Mar 2026 - Present

- Tutored 50+ students in debugging Python programs, diagnosing faults in unfamiliar code under time pressure and explaining the fix so it was understood rather than just applied.

Peer Mentor RMIT University | Jan 2025 - Present

- Mentor students across five subjects, including three I achieved High Distinctions in (Programming Studio 1, Algorithms and Analysis, Cloud Developing), plus Foundations of AI and Introduction to Mathematics for Computing.

Web Developer Mascot Enterprise (family business)

- Designed, built and deployed a React and Node.js site for my family's machinery parts export business, serving buyers in Asia, Europe and West Africa (mascot-business.netlify.app).

Food and Beverage Attendant, Delaware North (MCG, Marvel Stadium, VRC) May 2025 - Present

- Recognised at the MCG for hard work and dedication across the 2026 AFL season; high-volume event-day service with POS, EFTPOS and cash handling. Also crew at Guzman y Gomez and Domino's.

PROJECTS

Cancer Cell Classification from Histopathology Images (High Distinction) Python, CNNs | 2026

- Classified 20,000+ colon cell images from 99 patients as cancerous or not and into four cell types, splitting by patient so every test cell came from an unseen patient; built up from logistic regression and MLP to CNNs with targeted augmentation and class weighting; 88% accuracy on unseen patients.

Campus Hub, Event Management Platform (six-person team) Java, Spring Boot, React | Jul 2025 - Nov 2025

- Coordinated delivery (client and team meetings, minutes, unblocking slipping tasks) and owned RSVP notifications, cron-scheduled reminder emails, the event gallery and home page; presented in a live client demo and Q&A.
- REST APIs in Spring Boot on MySQL with a React front end, RBAC across three user types with an organiser approval workflow, Docker, Flyway migrations, unit tests and a GitHub Actions CI/CD pipeline.

Persuasion Technique Detection in Memes PyTorch, RoBERTa | 2026

- Multi-label classifier and span tagger for 20 persuasion techniques on 750 examples; compared frozen-encoder and fine-tuning strategies, implemented LoRA from scratch and a custom label-attention head, trained on an AWS SageMaker T4 GPU.

Retrieval-Augmented Survey Estimation (Kaggle) Python, RAG | 2026

- BM25 and dense retrieval with pseudo-relevance feedback, fused by reciprocal rank fusion; LLM-as-judge evidence scoring aggregated into answer distributions within a USD $3 API budget, with per-run token logging.

AWS Cloud Operations Project AWS | 2026

- Multi-AZ custom VPC with a bastion host, auto-scaled web tier behind an Application Load Balancer, database, Parameter Store and S3, monitoring and security controls, and a CloudFormation break-fix, all verified through the AWS CLI.

Emotion-Adaptive NPC, Research Collaboration with Justin Pierre RMIT | Jul 2025

- Tested a game NPC that used facial expressions, voice tone and sentiment analysis to decide whether to stay engaged; gave structured feedback.

Also: Pac-Man reinforcement learning agents (value iteration, Q-learning; 25/25), a Prolog crafting planner, a Java/Javalin food-loss web app, and a C++ Minecraft maze generator and solver with a self-written Bash test harness.

AVAILABILITY

Available from November 2026; no notice period required.